Problem Set 4 – The Distribution of Childcare Prices

MGT-403 Data Project · Where Should We Open Our Next Daycare?

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Summary of answers

- **Q1 sample.** NDCP county prices for 2015–2022, restricted to counties whose OEWS area code begins with “00” (metro). This leaves 1,235 metro counties in 2022, of which 1,013 report a center-based preschool price.
- **Q2 (2022).** Average county median weekly center-based price: infant $214.95, toddler $186.46, preschool $175.09. The 95% confidence intervals do not overlap, so care gets significantly cheaper as children get older.
- **Q3 (real trends).** Nominal prices rose about 22–24% from 2015 to 2022, almost exactly the CPI-U's 23.5%. Real prices rose modestly through 2020, then fell back in 2021–2022 as general inflation outpaced childcare price growth. In real terms, 2022 prices are roughly back at their 2015 level.
- **Q4 (CCDF voucher test).** 2022: $t = 6.39$, $p = 2.54 \times 10^{-10}$, so we reject H_0. The average preschool price ($175.09/wk) exceeds the voucher ($163.05/wk). 2015: $t = 23.32$, $p = 1.84 \times 10^{-96}$, a much larger shortfall. The voucher has closed most of the gap but still falls short of the average price.

Q1. Data and the metro analysis sample

Data. U.S. Department of Labor National Database of Childcare Prices, county-level workbook *NDCP2022.xlsx* (2008–2022; 2015–2022 used). The price variables are the median full-time weekly prices for **center-based** care, in nominal dollars: **MCINFANT**, **MCTODDLER** and **MCPRESCHOOL**.

Metro sample. The workbook stores FIPS codes as integers, so I restored the 5-character zero-padded code (e.g., 1001 → 01001). I then merged on [county FIPS, year] with the course crosswalk *county_oews_crosswalk.csv*, reading its codes as text so leading zeros are kept. All 25,763 NDCP county-years for 2015–2022 matched the crosswalk (25,763 matched). I kept counties whose 7-character *oews_area* begins with “00” (metropolitan) and dropped the nonmetropolitan codes. This leaves 9,873 metro county-years.

Year	Metro counties	with infant price	with toddler price	with preschool price
2015	1,233	1,003	1,003	1,003
2016	1,233	1,025	1,025	1,025
2017	1,234	1,063	1,063	1,063
2018	1,233	1,062	1,063	1,063
2019	1,235	1,062	1,063	1,063
2020	1,235	1,062	1,063	1,063
2021	1,235	916	916	918
2022	1,235	1,012	1,011	1,013

Table 1. Metro analysis sample by year. Counties without a price for an age group are left out of that age group's statistics.

Q2. 2022 distribution of center-based prices by age group

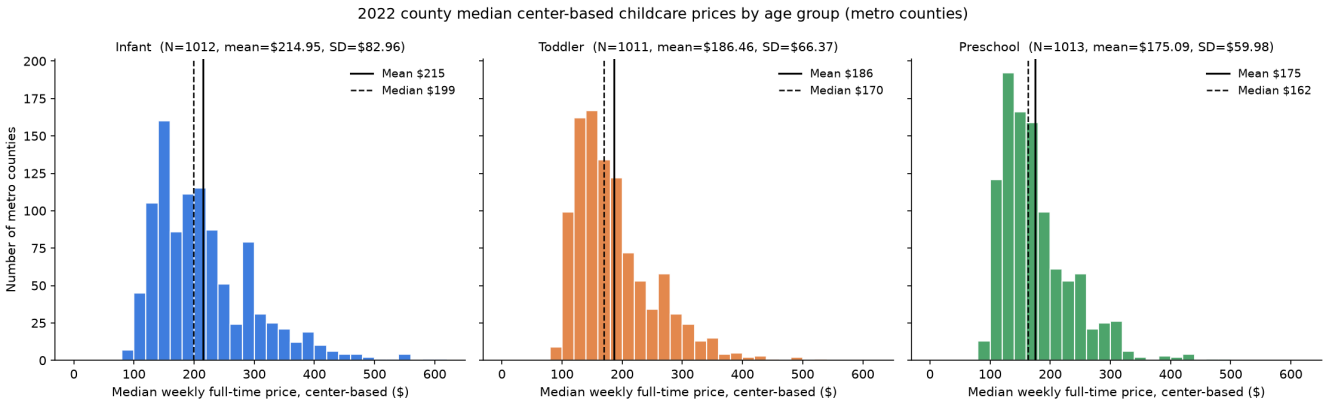


Figure 1. Histograms of 2022 county median weekly center-based prices (metro counties, $20 bins). Solid line = mean across counties; dashed line = median across counties.

Age group	N	Mean	SD	SE = SD/√N	$t_{0.975,\,N-1}$	95% CI for mean	Median	Min	Max
Infant	1012	$214.95	$82.96	$2.608	1.9623	[$209.84, $220.07]	$198.93	$95.60	$606.61
Toddler	1011	$186.46	$66.37	$2.087	1.9623	[$182.36, $190.55]	$170.00	$97.00	$520.00
Preschool	1013	$175.09	$59.98	$1.885	1.9623	[$171.39, $178.79]	$162.50	$86.19	$496.30

Table 2. 2022 county median weekly center-based prices (nominal = 2022 dollars). 95% CI = mean ± $t_{0.975,\,N-1}$ × SD/√N.

Discussion

- **Care gets cheaper as children get older.** Mean infant care ($214.95) costs $28.53/wk (15.3%) more than toddler care ($186.46), and toddler care costs $11.31/wk (6.5%) more than preschool ($175.09). Infant care averages 22.7% more than preschool.
- **The differences are statistically significant.** None of the three 95% CIs overlap, so the means differ at the 5% level. Among the 1,011 counties reporting all three prices, infant care costs more than preschool in 96.4% of counties (paired t = 34.4). Infant care costs more than toddler care in 92.4% (paired t = 29.5). The toddler–preschool gap is smaller: toddler care costs more in 64.5% of counties and exactly the same in 28.6% (paired t = 19.0).
- **Prices vary more for younger children.** The SD falls from $82.96 (infant) to $66.37 (toddler) and $59.98 (preschool).
- **The distributions are right-skewed.** In every age group the mean exceeds the median (e.g., preschool $175.09 vs. $162.50), and there is a long upper tail of high-price counties (preschool max $496.30/wk). The mean is therefore pulled above the typical county; this matters for Q4.
- *Interpretation (not tested with this data):* a likely explanation for the age pattern is that younger children need more staff per child, and labor is one of the two main costs of providing care (labor and rent, as the assignment notes).

Q3. Real (constant 2022 $) price trends, 2015–2022

Method. Real price$_t$ = nominal price$_t$ × CPI$_{2022}$ / CPI$_t$, using CPI-U (series CUUR0000SA0) annual averages from the BLS API. For each year and age group I plot the mean of the county median prices across metro counties, with a 95% CI. The set of reporting counties changes from year to year (Table 1), so I also plot a **balanced panel**: only counties that report a price in all eight years. Those lines show price changes that aren't driven by which counties report.

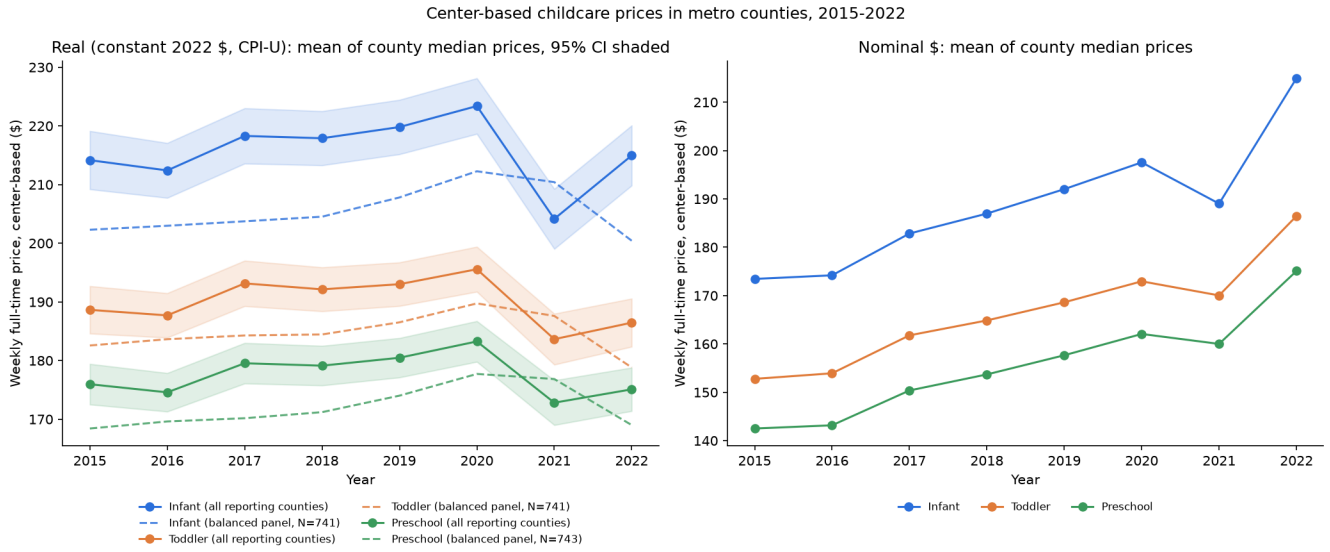


Figure 2. Left: real 2022-dollar prices (solid = all reporting counties with 95% CI; dashed = balanced panel). Right: the same all-county means in nominal dollars.

Year	CPI-U	Infant	Toddler	Preschool	Infant (bal.)	Toddler (bal.)	Preschool (bal.)
2015	237.017	$214.16	$188.64	$175.96	$202.30	$182.56	$168.41
2016	240.007	$212.40	$187.69	$174.58	$202.97	$183.63	$169.61
2017	245.120	$218.29	$193.15	$179.54	$203.74	$184.27	$170.17
2018	251.107	$217.89	$192.14	$179.12	$204.52	$184.45	$171.20
2019	255.657	$219.80	$193.01	$180.47	$207.80	$186.51	$174.01
2020	258.811	$223.38	$195.56	$183.25	$212.26	$189.74	$177.71
2021	270.970	$204.14	$183.64	$172.81	$210.41	$187.60	$176.85
2022	292.655	$214.95	$186.46	$175.09	$200.42	$178.82	$168.99

Table 3. Mean county median weekly price, real 2022 dollars. “bal.” = balanced panel (N = 741 infant, 741 toddler, 743 preschool counties).

Age group	Nominal 2015	Nominal 2022	Nominal chg.	Real chg. (all)	Real chg. (balanced)	Balanced real 2015→2020	Balanced real 2020→2022
Infant	$173.44	$214.95	+23.9%	+0.4%	-0.9%	+4.9%	-5.6%
Toddler	$152.78	$186.46	+22.0%	-1.2%	-2.1%	+3.9%	-5.8%
Preschool	$142.51	$175.09	+22.9%	-0.5%	+0.3%	+5.5%	-4.9%

Table 4. Changes over the period. CPI-U inflation: 2015→2022 +23.5%; 2020→2021 +4.7%; 2021→2022 +8.0%.

Discussion: what happened to childcare prices?

- **Nominal prices rose a lot, but only about as fast as overall inflation.** Mean nominal prices rose 23.9% (infant), 22.0% (toddler) and 22.9% (preschool), while the CPI-U rose 23.5%. In constant 2022 dollars, prices in

2022 are within about 2% of their 2015 level in every age group, both in the full sample and in the balanced panel.

- **Real prices rose until 2020, then fell back.** In the balanced panel, real preschool prices rose steadily from $168.41 (2015) to $177.71 (2020), up 5.5%. They then fell to $176.85 in 2021 and $168.99 in 2022. Infant and toddler prices follow the same pattern. For these same counties, nominal prices kept rising in 2021–22, but more slowly than the post-pandemic inflation surge (CPI +4.7% in 2021, +8.0% in 2022), so their real value fell.
- **The sharp 2021 dip in the all-county line is partly caused by which counties report.** In 2021, several states have no metro price data: CA (37), MO (7), NC (46), PA (37), VA (80) counties in 2020. Meanwhile Puerto Rico appears for the only time in 2015–2022: PR (69 municipios, mean $78.12/wk nominal). The counties that dropped out averaged $170.65/wk in 2020 (nominal), above the 2020 all-county mean of $162.06. Replacing them with PR's very low-priced municipios pulls the 2021 all-county mean down. The balanced panel, which holds the counties fixed, shows only a small 2021 decline (-0.5% for preschool).
- **The ranking by age never changes:** infant > toddler > preschool in every year.

Q4. Does the CCDF voucher equal the price of median center-based preschool?

Setup. The voucher is monthly; NDCP prices are weekly. With 4.33 weeks per month, the weekly voucher value is $706 / 4.33 = **$163.0485** in 2022 and $473 / 4.33 = **$109.2379** in 2015 (nominal). Each year's voucher is compared with *that year's nominal* prices, so no inflation adjustment is needed. “The price of care at the median-priced center” is the NDCP variable **MCPRESCHOOL** (each county's median center-based preschool price). “The average metro area” is the population mean of that variable across metro-area counties (the Q1 sample).

(a) Hypotheses

Let μ = mean county median weekly center-based preschool price across metro counties in 2022.

H_0: μ = \$163.05 (the voucher equals the average price) H_1: μ ≠ \$163.05 (two-sided, α = 0.05)

The test is two-sided because the claim is that the voucher *equals* the price; a voucher that is too high or too low would both contradict it. Multiplying through by 4.33 gives the equivalent monthly statement, H_0: 4.33μ = \$706; the t-statistic is the same either way.

(b) Test statistic, computed by hand (2022)

$$t = (\bar{x} - \mu_0) / (s / \sqrt{n}), \quad \text{distributed } t_{n-1} \text{ under } H_0$$

Element	Definition	2022 value
n	number of metro counties with a preschool price	1,013
$\bar{x}$	sample mean of MCPRESCHOOL	\$175.0885
s	sample standard deviation (n − 1 denominator)	\$59.9800
√n	square root of n	31.8277
SE = s / √n	standard error of the mean	59.9800 / 31.8277 = 1.8845
μ_0	706 / 4.33	\$163.0485
$\bar{x} - \mu_0$	gap between average price and voucher (per week)	175.0885 − 163.0485 = 12.0400
t	$(\bar{x} - \mu_0)$ / SE	**12.0400 / 1.8845 = 6.3889**
df	n − 1	1,012

(c) Software check and p-value

Python / SciPy computed the same test directly from the data (`scipy.stats.ttest_1samp(x, 706/4.33)`, plus the hand formula in `ps4.py`):

$$\text{p-value} = 2 \times P(T_{n-1} \geq |t|) = 2 \times [1 - F_{t,\ 1012}(6.3889)] = \mathbf{2.54 \times 10^{-10}}$$

SciPy gives t = 6.3889 and p = 2.54×10^{-10}, identical to the manual t = 6.3889, **so the hand calculation is verified**. I also re-read the raw workbook with a separate program (openpyxl plus hand-coded formulas) and got the same n, mean, SD, and t.

(d) Interpretation, and the same test for 2015

	2022	2015
Monthly voucher (nominal)	$706	$473
μ_0 = voucher / 4.33 (weekly)	$163.05	$109.24
n (metro counties)	1,013	1,003
$\bar{x}$ (mean weekly price)	$175.09	$142.51
s	$59.98	$45.17
SE	1.8845	1.4264
t statistic (df)	6.389 (1,012)	23.324 (1,002)
p-value (two-sided)	2.54×10^{-10}	1.84×10^{-96}
Critical |t| at 5%	1.962	1.962
Decision	Reject H_0	Reject H_0
Mean price per month ($\bar{x} \times 4.33$)	$758.13	$617.06
Monthly shortfall (price − voucher)	$52.13	$144.06
Voucher as % of mean price	93.1%	76.7%
Share of counties priced above voucher	49.0%	75.1%
Median county price (weekly)	$162.50	$129.28

Table 5. One-sample t-tests of the ACF claim, 2022 and 2015 (nominal dollars).

- **2022: reject H_0.** |t| = 6.39 is far above the 5% critical value of 1.96, and p = 2.54×10^{-10} is far below 0.05. If the voucher truly equaled the average price, a sample mean this far from it would almost never happen by chance. The average metro county's median preschool price is $175.09/week ($758.13/month). That is $12.04/week, or $52.13/month, more than the voucher, so the voucher covers about 93% of the average price. The ACF's claim is not supported: the voucher is significantly *below* the average price.
- **2015: reject H_0 even more strongly** (t = 23.32, p = 1.84×10^{-96}). The 2015 voucher was $144.06/month short of the average price and covered only about 77% of it; 75% of metro counties were priced above it. The voucher rose 49.3% in nominal terms from 2015 to 2022, compared with 22.9% for average prices. This closed most of the gap, but not all of it.
- **Mean vs. median.** In 2022 the *median* county's price ($162.50) is within about a dollar of the weekly voucher, and 49% of counties are priced above it. So the voucher roughly matches the price in the *typical* (median) metro county. It is below the *average* because the right-skewed distribution (Q2) pulls the mean up. The claim as stated is about the average metro area, and that is what the test rejects.
- **Robustness check.** Counting each OEWS metro area once (averaging its counties) gives the same conclusion: 2022 t = 4.33, p = 1.96×10^{-5} (341 metro areas, mean $175.33); 2015 t = 15.59, p = 9.93×10^{-42}.

Assumptions and data-handling choices

- Unit of observation: metro county (NDCP is county-level; the assignment asks for a consistent county set across PS4, PS5, and the project). Each county counts equally; the statistics are not population-weighted.
- “Median price” = NDCP median (not 75th percentile) full-time weekly price for center-based care: MCINFANT, MCTODDLER, MCPRESCHOOL. Family/home-based care (MFCC*) is excluded.
- Counties with a missing price are dropped only for that age group and year; N differs slightly across age groups (Table 1).
- 95% CIs use the t distribution ($t_{0.975,\,N-1} \approx 1.962$); with N > 1,000, a normal approximation (1.96) gives CIs within one cent of these.
- Puerto Rico's metro municipios are in the crosswalk's metro sample, but NDCP reports their prices only in 2021 (so they do not affect the 2022 or 2015 results).
- CPI-U: series CUUR0000SA0, annual averages, fetched from the BLS Public Data API (saved to data/cpi_u_annual.csv).
- Reproduce: put NDCP2022.xlsx and county_oews_crosswalk.csv in data/, then run `python ps4.py` and `python make_report.py`. Every number in this report is read from output/results.json, which ps4.py writes.